

Faculty Of Engineering and Technology

Electrical and Computer Engineering Department

ENEE 4113, Communication Laboratory

Normal Amplitude Modulation and Demodulation

Report#1

Experiment #1

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Abstract

The aim of this experiment is to study the principles and behaviour of Amplitude Modulation (AM) and its demodulation techniques in both the time and frequency domains. The experiment focuses on understanding how changes in the message signal's amplitude and frequency affect the modulation index and the overall performance of the AM system. Coherent and non-coherent demodulation methods are analysed to compare their effectiveness in recovering the original message signal. Additionally, frequency domain analysis using FFT is performed to observe the spectral components of the signals at various stages. This experiment provides an understanding of AM transmission and reception, emphasizing the impact of system parameters on signal quality and recovery.

Theory:

What is Modulation?

Modulation is the process of varying a property of a periodic waveform, such as amplitude, frequency, or phase, to encode information onto a carrier signal for transmission. This technique is fundamental to telecommunications, allowing information like sound, images, and data to be sent over various media like radio waves or telephone lines. Common examples include amplitude modulation (AM) and frequency modulation (FM) used in broadcasting

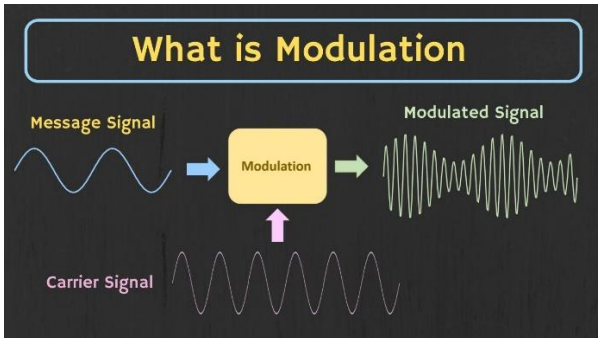


Figure 1 Modulation

Normal Amplitude Modulation:

Amplitude Modulation is a technique by which the amplitude of the carrier wave is changed according to the signal wave or modulating signal. Among various modulation schemes, amplitude modulation is the simplest and oldest modulation technique.

Basically, the information that is carried by the low frequency modulating signal is superimposed on the carrier wave of high frequency by varying the amplitude of the carrier. Thus, it is termed as amplitude modulation.

Mathematically the message signal $m(t) = Am\cos(2\pi fmt)$ and the carrier signal $c(t) = Ac\cos(2\pi fct)$ combine to form the modulated signal s(t)=[1+km(t)]c(t) which k Is the modulation sensitivity $k_a = \mu/Am$

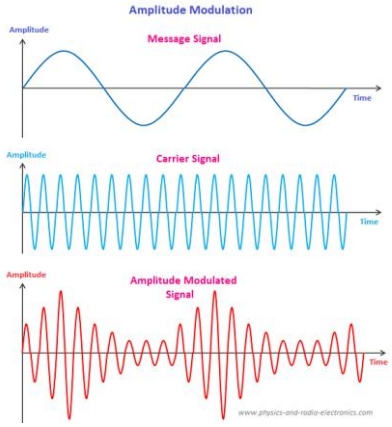


Figure 2 Amplitude Modulation

Modulation index:

The modulation index (μ) in AM is the ratio of the modulating signal's peak amplitude Am to the carrier's peak amplitude Ac $\mu = \frac{Am}{Ac}$ Alternatively, it can also be calculated from the modulated waveform using the envelope $\mu = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$

It indicates the extent of amplitude variation in the modulated carrier. A modulation index of 1 (or 100%) represents maximum modulation without distortion, while a value greater than 1 result in overmodulation, causing the carrier to be distorted.

The modulated signal using μ = s(t)=Ac [1+m cos(ωmt)] cos(ωct)

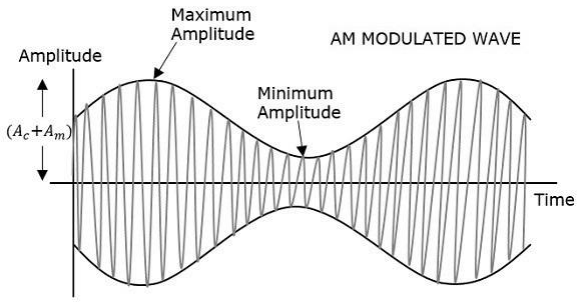


Figure 3 Normal Am Modulation

Types of Modulation in Normal Am:

- μ=1

(100% modulation): The modulation is optimal, providing maximum power to the sidebands without causing distortion. In this case, the amplitude of the signal varies but does not become zero.

- μ<1

(Under-modulation): The carrier's amplitude is varied by less than its maximum possible extent. This is a normal operating condition but uses power less efficiently.

- μ>1

(Over-modulation): This occurs when the modulating signal's amplitude is greater than the carrier's amplitude, causing the carrier's envelope to dip to zero and create phase reversals. This results in distortion and can cause interference.

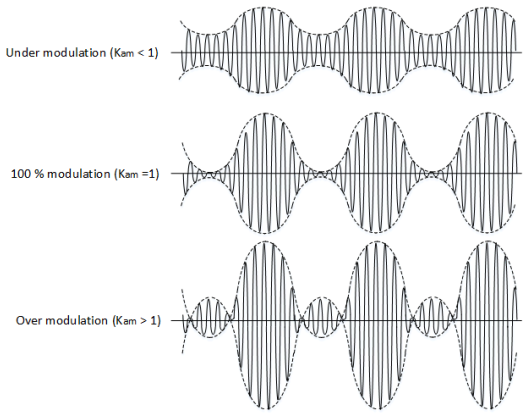


Figure 4 Types of modulation

Power Efficiency in AM:

The power efficiency of a normal AM signal is defined as:

$$\eta = \frac{power\ in\ the\ sidebands}{power\ in\ the\ sidebands + power\ in\ the\ carrier}$$

$$\text{Power in carrier} = \frac{A_C^{\,2}}{2}$$

$$\text{Power in sidebands} = \frac{1}{2}\left(\frac{A_C\mu}{2}\right)^2 + \frac{1}{2}\left(\frac{A_C\mu}{2}\right)^2 = \frac{1}{4}A_C^{\,2}\mu^2$$

$$\text{Therefore,} \quad \eta = \frac{\frac{1}{4}A_C^{\,2}\mu^2}{\frac{A_C^{\,2}}{2} + \frac{1}{4}A_C^{\,2}\mu^2} = \frac{\mu^2}{2+\mu^2} \quad ; \quad 1 \geq \mu \geq 0$$

Figure 5 Power and Power efficiency

which is at best 1/3 when μ=1. So, 2/3 of the energy goes to the carrier as losses energy.

Demodulation:

The Demodulation is the process of extracting the original message signal m(t) from a modulated Signal s(t). It is an important function used in the communication systems which allows the recovery of transmitted data at the receiver end. There are 3 types of demodulations techniques: envelope detector demodulation, coherent demodulation and non-coherent demodulation.

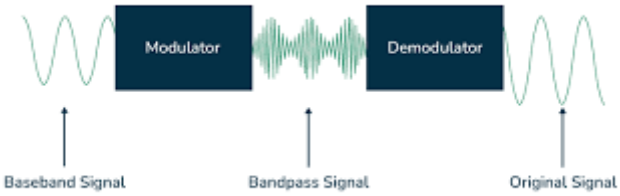


Figure 6 Modulation and demodulation

Types of Demodulations:

1. Envelope Detection:

   Envelope detection is a simple and efficient method for demodulating AM signals. It works by extracting the envelope of the modulated waveform, which corresponds to the original message signal m(t). Conditions for using an envelope detector :

   $$f_c > 10 f_m$$

   Modulation Index between $0 \leq \mu \leq 1$

   $$1/f_c \ll RC \ll 1/f_m$$

   (RC is the time constant of the RC low pass filter)

   Figure 7 Envelope detector conditions

   so, the carrier frequency must be significantly higher than the message frequency to ensure accurate tracking of the envelope.

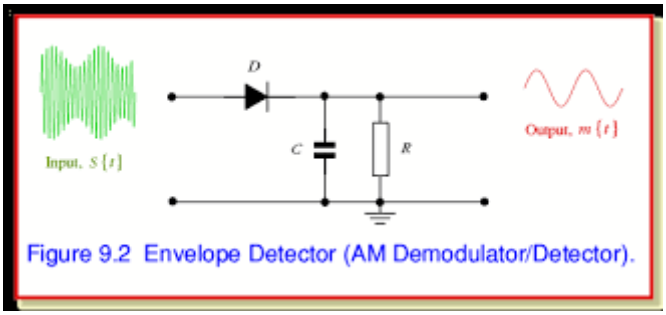


Figure 8 Envelope Detector circuit

2. Coherent demodulation:

   is a method used to recover the original message signal m(t) from a modulated signal s(t) by multiplying it with a synchronized carrier signal at the receiver. This technique requires that the receiver generate a carrier wave that is identical in frequency and phase to the one used in the transmitter. After multiplication, the signal passes through a low-pass filter (LPF) to remove high-frequency components, leaving the original message signal.

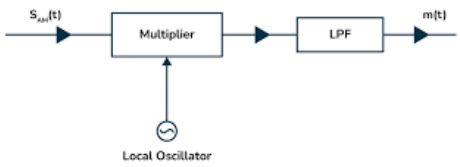


Figure 9 Coherent demodulation circuit

3. Non – Coherent demodulation:

is a method used to recover the message signal m(t) from the modulated signal s(t) without needing a carrier that is in sync with the original one used during transmission. It does not require the receiver to match the exact frequency or phase of the transmitter's carrier, making it easier to implement. This method is commonly used in basic AM receivers.

Procedure

Part 1: Normal Amplitude Modulation

In this experiment, the sinusoidal message signal is modulated using Amplitude Modulation (AM) with the following parameters:

- Peak-to-peak amplitude of the message signal (Vss) is 4 volts.
- Message signal frequency (fm) is 2 kHz.

Message signal in time domain and frequency domain

1. Message signal in time domain

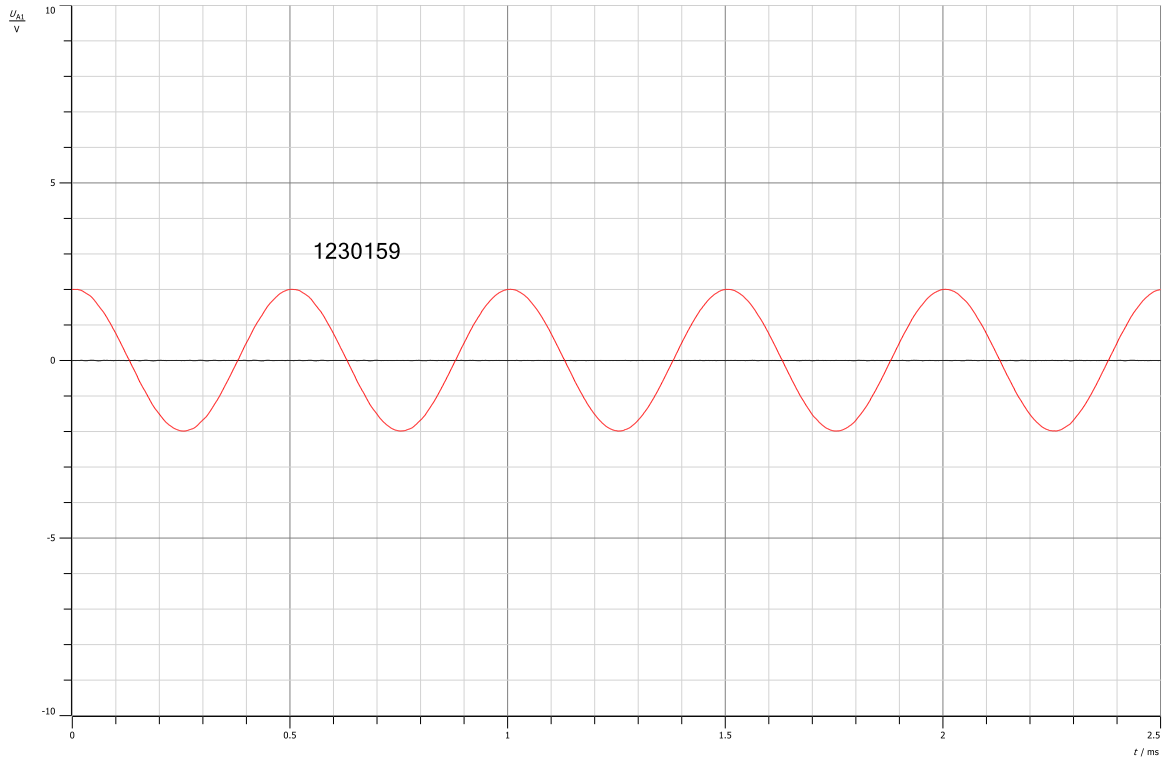


Figure 10 Message signal m(t) in time domain

2. Message signal in frequency domain

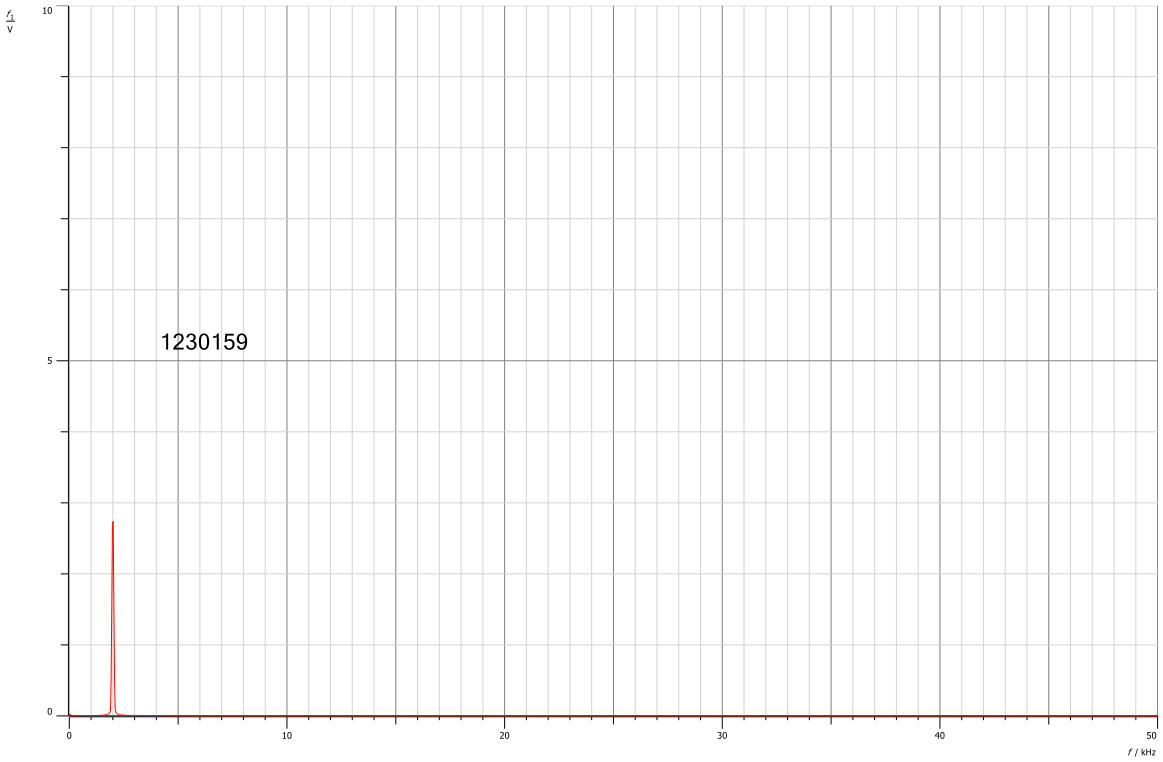


Figure 11 Message signal m(t) in frequency domain

in the frequency domain (after Fourier Transform), this signal produces two impulses:

$$X(f) = \frac{A}{2}[\delta(f - f_0) + \delta(f + f_0)]$$

That means:

- One impulse at $+f_0$ ====> +2KHz
- One impulse at $-f_0$====> -2KHz
- Each impulse has amplitude $Am/2$ ====> 2

However, the cosine and sine waves have the same frequency spectrum in magnitude, but they different in phase.

Carrier Signal in time and frequency domain:

After this the carrier in this experiment is generated using a pulse frequency oscillator of 160kHz. After that it is entered to a frequency division block to out a 20kHz pulse signal.

1. Carrier Signal in time domain

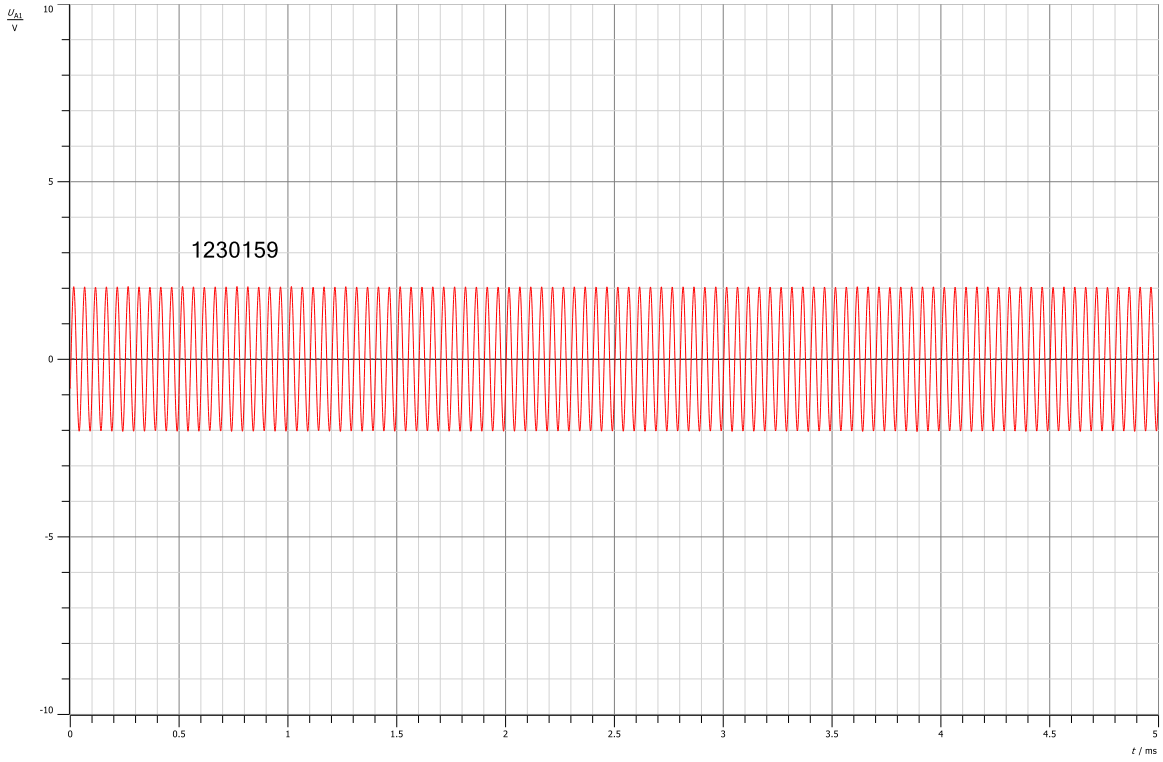


Figure 12 Carrier signal c(t) in time domain

Carrier amplitude Ac = 2

Carrier Frequency Fc = 20KHz

2. Carrier Signal in Frequency domain

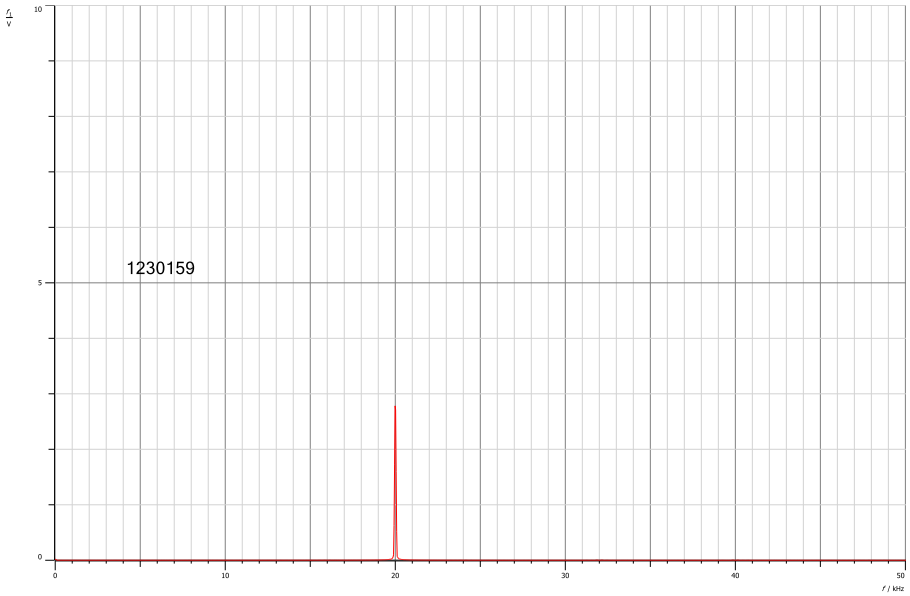


Figure 13 Carrier signal c(t) in frequency domain

Modulated Signal in time and frequency domain:

The modulated signal s(t) is generated as follows:

$$s(t) = (1 + ka\, m(t))\, c(t) = c(t) + k\, m(t)\, (t)$$

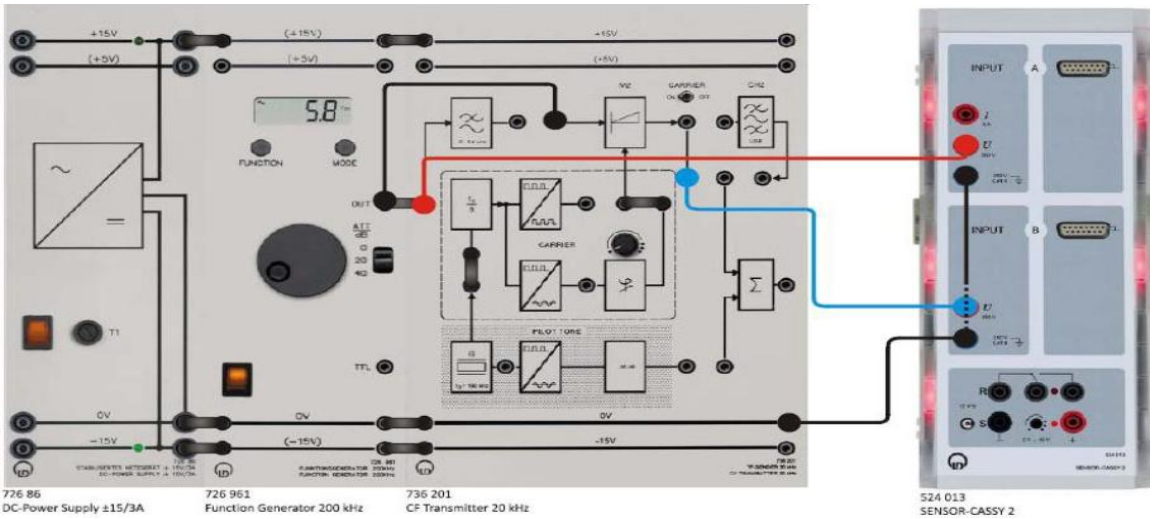


Figure 14 Connection for s(t) in the lab

1. Modulated signal in time domain

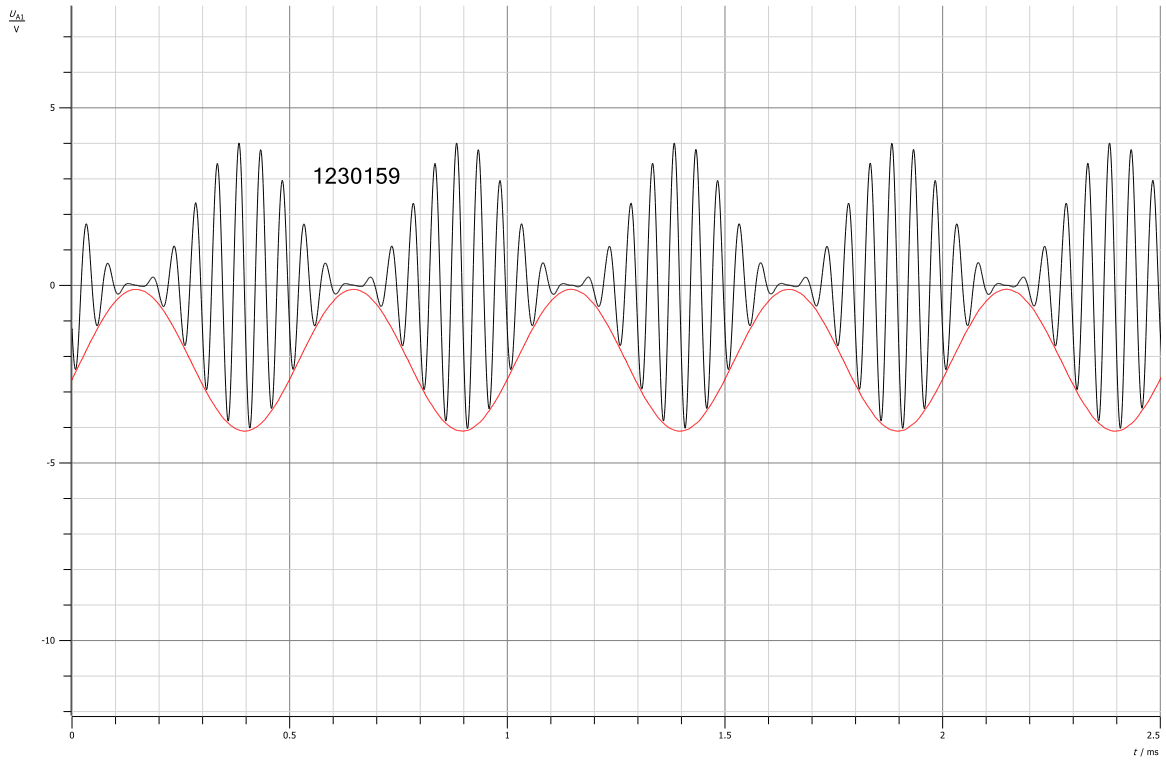


Figure 15 Modulated signal s(t) in time domain

The message signal, m(t), is the original low-frequency signal that carries the information to be transmitted. In amplitude modulation (AM), this message signal modulates the amplitude of a high-frequency carrier wave, resulting in the modulated signal s(t).

In the time domain, m(t) acts as the envelope of the modulated signal s(t). This means that the amplitude of s(t) follows the shape of m(t). When the message signal reaches its peak, the amplitude of the modulated signal also reaches its maximum. While, when the message signal is at its minimum, the amplitude of the modulated signal is at its lowest.

Thus, the message signal controls the amplitude variations of the carrier wave and determines the envelope of the modulated signal.

Modulated Signal Expression:

$$\boxed{s(t) = [A_c + k_a\, m(t)]\cos\,(\omega_c t)}$$

2. Modulated Signal in Frequency Domain

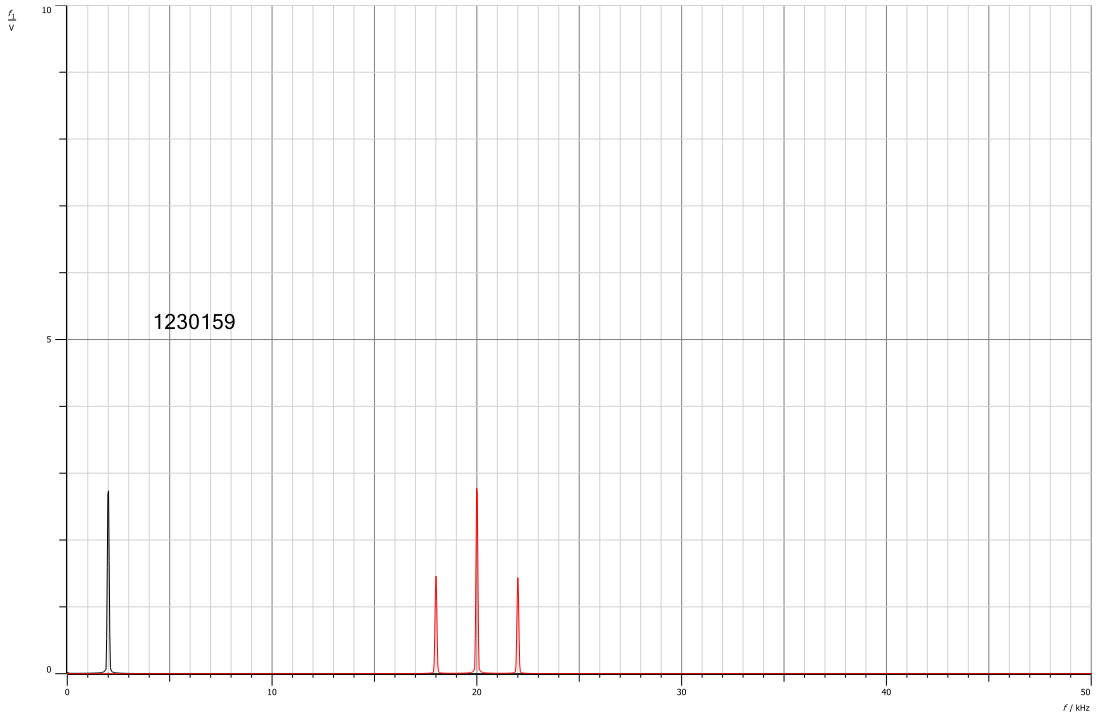


Figure 16 Modulated signal s(t) in frequency domain

As shown in the figure above the there are three impulses the central one at 20kHZ which is the Fc, and two-sided impulses one at 22kHZ and the second one at 18kHz. to be lower side "Fc - Fm", carrier frequency, and the upper side "Fc + Fm." On the other hand, the black impulse is the message signal in the frequency domain at FM= 2KHz

Calculations on modulated signal:

1. Bandwidth:

- From the spectrum:
    - Lower side frequency = 18 kHz
    - Upper side frequency = 22 kHz

So,

$$\text{Bandwidth} = 22\,\text{kHz} - 18\,\text{kHz} = 4\,\text{kHz}$$

That means:

$$\text{Bandwidth} = 2 \times f_m$$

where $f_m = 2$ kHz (the highest message frequency).

2. Modulation index:

We know that $Am = 2$ and $Ac = 2$ so we can calculate the Modulation index by using

$\mu = \frac{A_m}{A_c} = \frac{2}{2} = 1$

3. Modulation Sensitivity:

we can calculate the modulation sensitivity k_a using.

$\mu = k_a * (A_m)_{max}$

$1 = k_a * 2 \rightarrow k_a = 0.5$

4. Power and Power efficiency:

$Ac = 2, \mu = 1$

- Power of USP $= \left(\frac{Ac*M}{2}\right)^2 = 1$
- Power of LSP $= \left(\frac{Ac*M}{2}\right)^2 = 1$
- Power of carrier $= \frac{Ac^2}{2} = 2$
- Power efficiency $= \frac{P_{sides}}{P_{total}} = \frac{\mu^2}{\mu^2+2} = 1/3$
- Maximum power efficiency when μ = 1, efficiency = 33%,

  Only 33% of the total power carries information (in the sidebands). The remaining 67% is wasted in the carrier, which has no information content that's why AM is power-inefficient.

As shown in the calculation we can realise that change message or carrier frequency does not change the value of the modulation index because it depends on the amplitude. Same as the modulator sensitivity.

Changing parameters:

Now, the effect of changing the modulating signal's frequency or amplitude on the modulation process will be studied.

First case Frequency:

Keep Vss = 4V and change the message frequency to 1kHz and 3kHz

1KHz

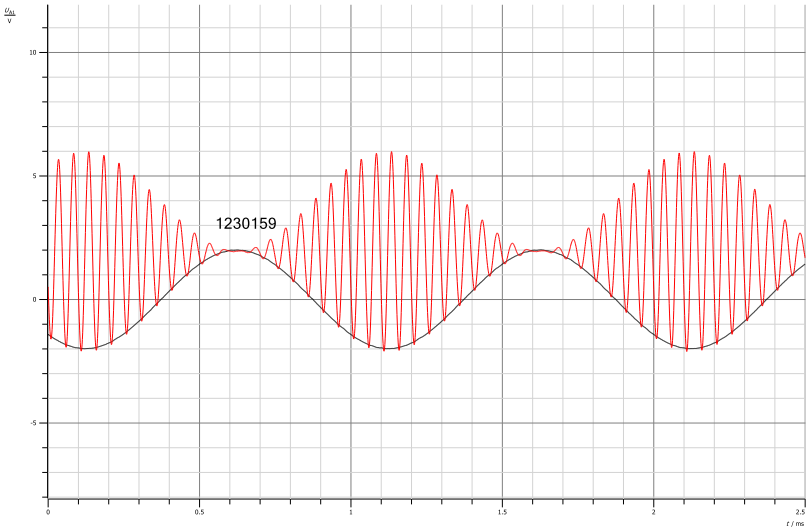


Figure 17 Modulated signal s(t) with 1KHz freq

3KHz

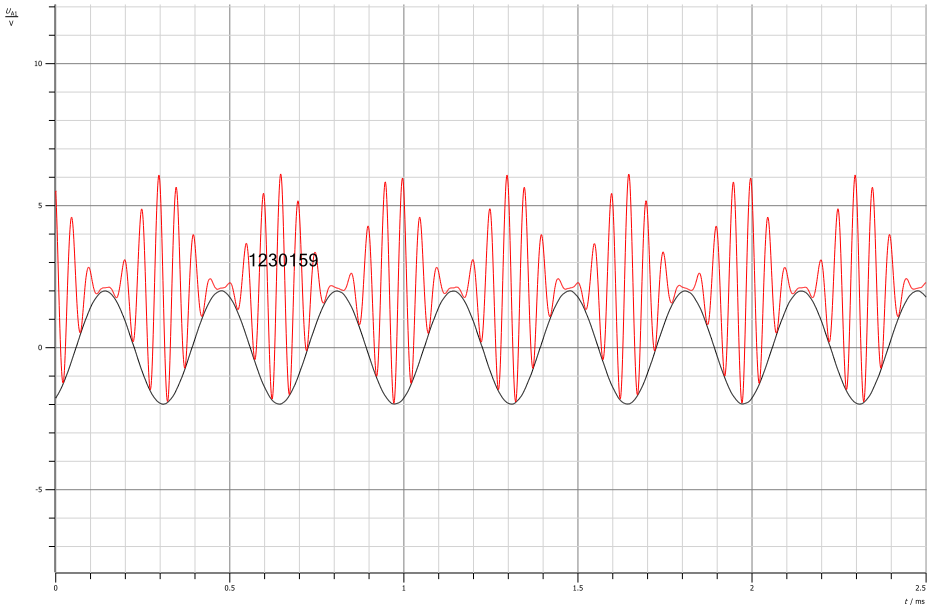


Figure 18 Modulated signal s(t) with 3KHz freq

Changing the message frequency has no effect on the modulation index, because the envelope amplitudes stay the same. So, the modulation index $m$does not depend on frequency, It depends only on amplitude.

Second Case Amplitude:

Keep fm = 2kHz and change the message Vss to 2V and 6V

Vss=2V

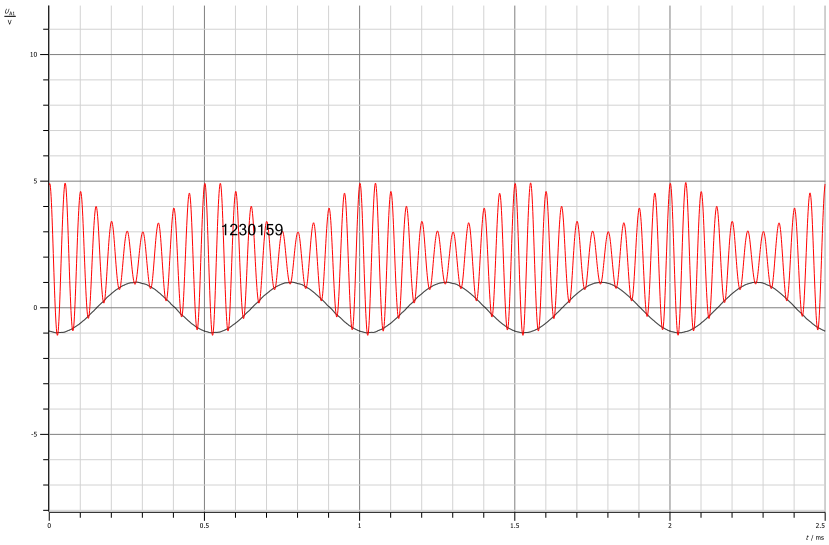


Figure 19 Modulated signal s(t) with Vss 2v

$A_m = \frac{Vss}{2}$

$\mu = k_a * (A_m)_{max}$ = 0.5 * 1 = 0.5 < ($\mu = 1$) Undermodulation

Vss= 6v

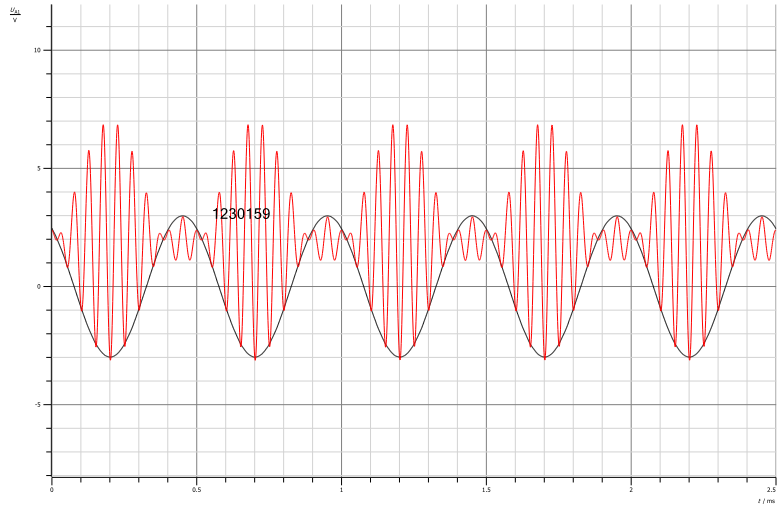


Figure 20 Modulated signal s(t) with Vss 6v

$\mu = k_a * (A_m)_{max}$ = 0.5 * 3= 1.5 > ($\mu = 1$) overmodulation

The envelope when the signal is over modulation not same as the message signal. So, in this case it is impossible to recover the signal using the envelope detector, "it's recovered when the method is coherent demodulation".

LPF phase shift effect:

In this section the requirements are feed the modulating signal m(t) via the low pass (LP) filter before the modulator. The first when the VSS = 4 volt and the frequency 3kHZ then change the frequency to 7kHZ.

First part (3KHz):

3KHz

Effect In time domain

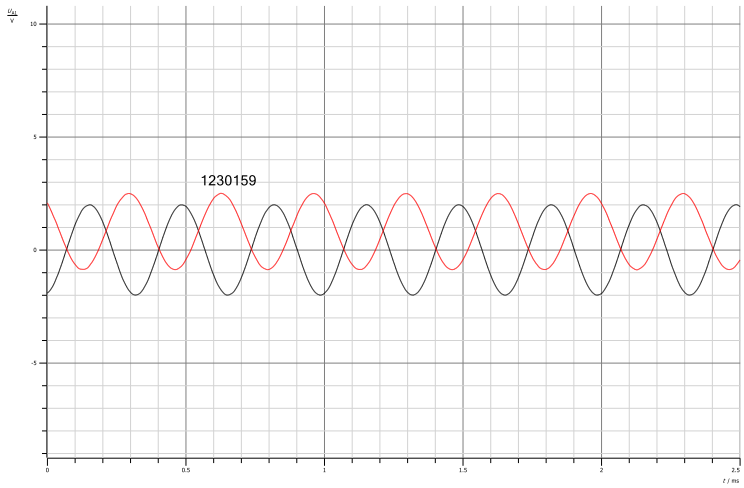


Figure 21 Message signal before and after the LPF when freq= 3khz in time domain

Effect In frequency domain

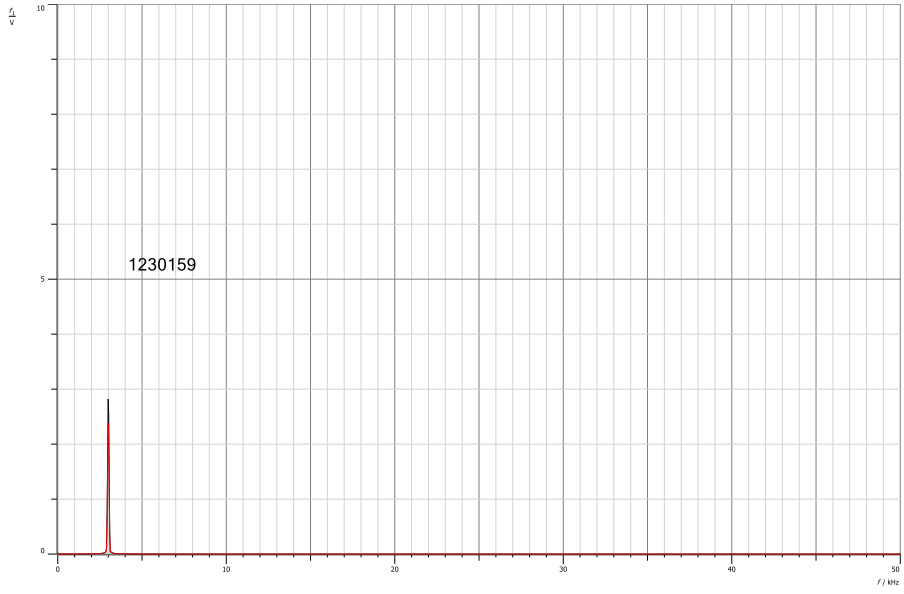


Figure 22 Message signal before and after the LPF in when freq = 3khz in freq domain

when we pass the signal into a LPF the signal shifted by 90 degrees.

Second part (7KHz):

7KHz

Effect in time domain

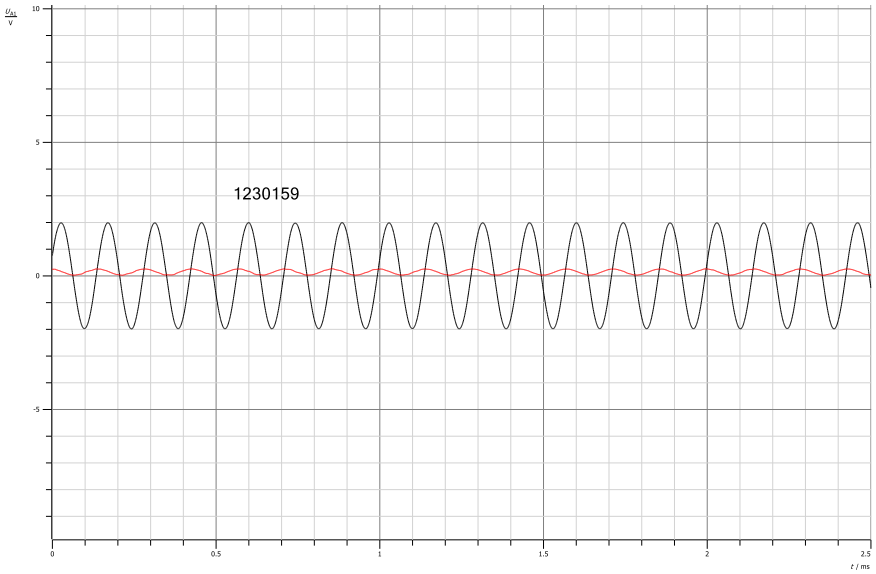


Figure 23 Message signal before and after the LPF, when frequency is 7Khz in time domain

Effect on frequency domain

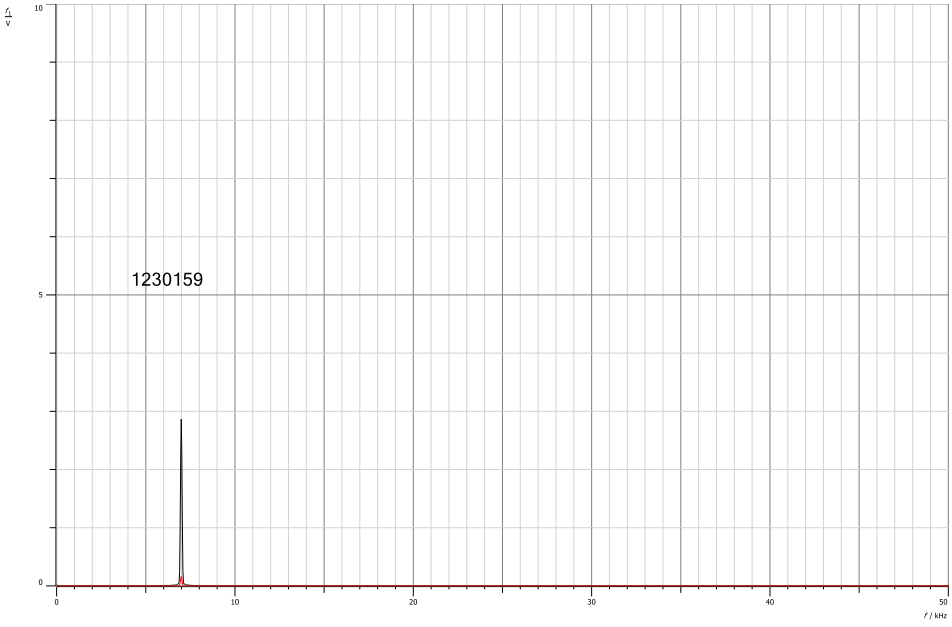


Figure 24 Message signal before and after the LPF, when frequency 7KHz in frequency domain

As shown in the plot, the low pass filter reduced the message signal because the message's frequency was outside of its range 3.4Khz, causing it to be cut off. And passing the message

signal through the LPF may also result in distortion of the signal due to phase shifts and delays introduced by the LPF.

Part 2: Normal Amplitude Demodulation:

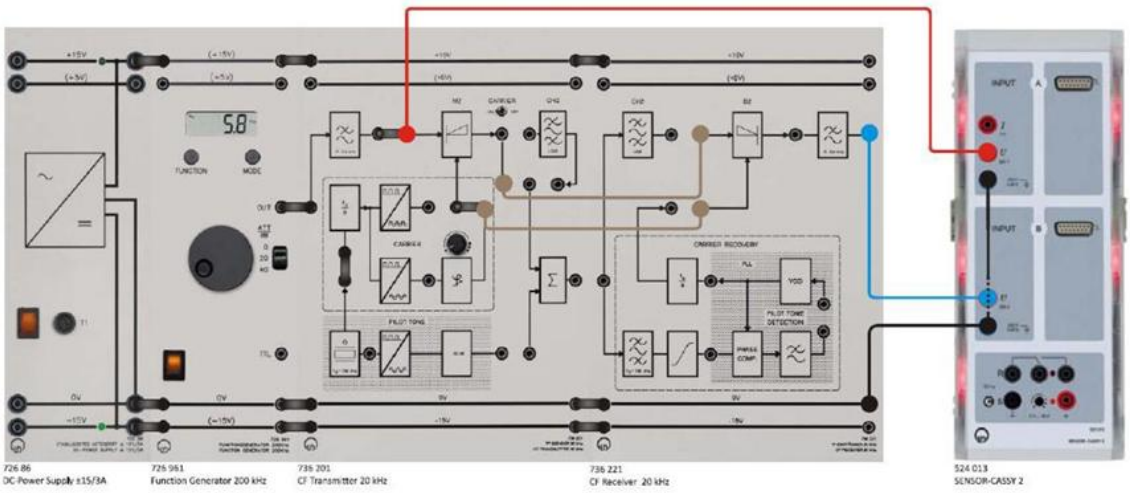


Figure 25 Connection for normal am demodulation in the lab

Coherent Demodulation

In normal AM (amplitude modulation) coherent demodulation is achieved by multiplying the modulated signal with a local oscillator signal that is in phase with the carrier signal.

The function generator is set to sine, VSS = 4V and fm = 2 kHz.

Output In time domain

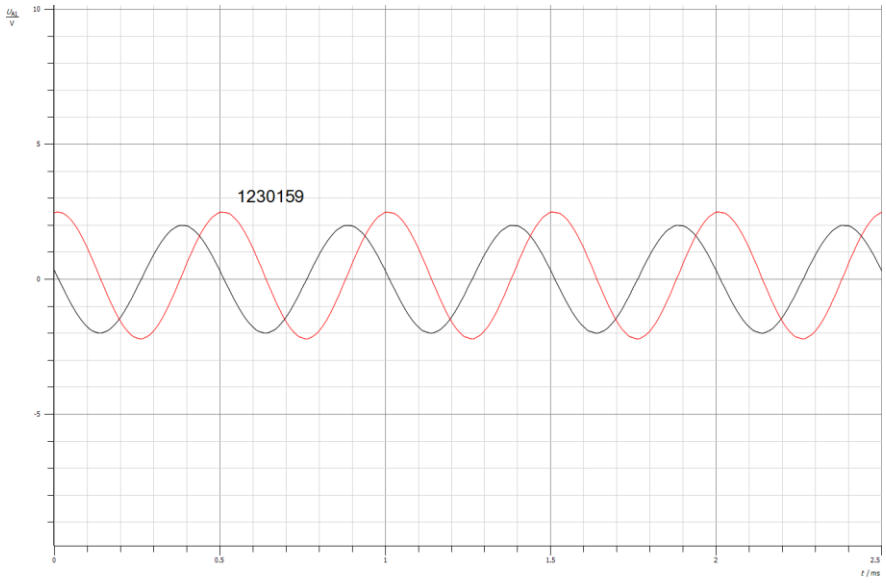


Figure 26 Coherent demodulation in time domain

Output in frequency domain

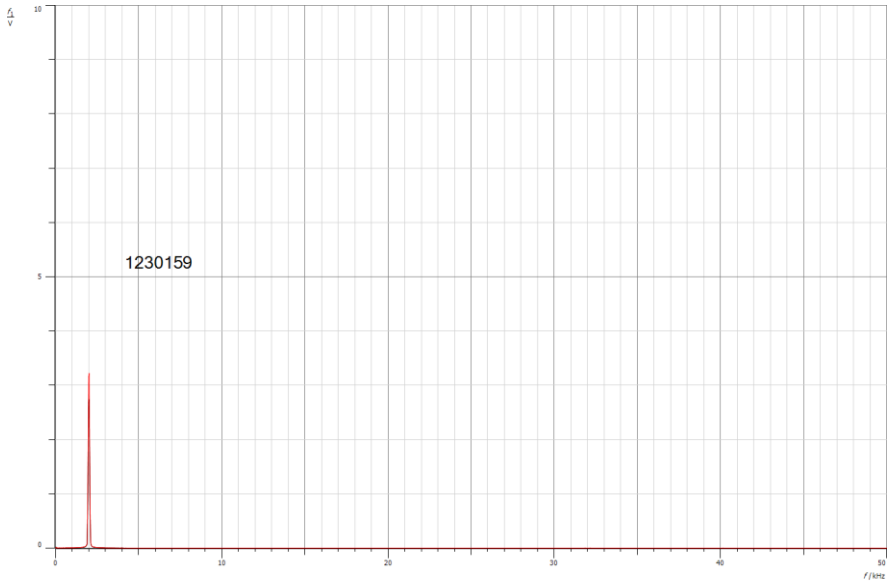


Figure 27 Coherent demodulation in frequency domain

After entering the demodulation signal result to the filter, the original message signal will appear with a shift (this effect of low pass filter)

Non-Coherent Demodulation

the effect of non-coherence, feed the modulator with a sinusoidal carrier from the pin before the phase shifter and feed the demodulator with the carrier from the pin after the phase shifter. Change the carrier phase from the phase shifter (φ).

Output in time domain

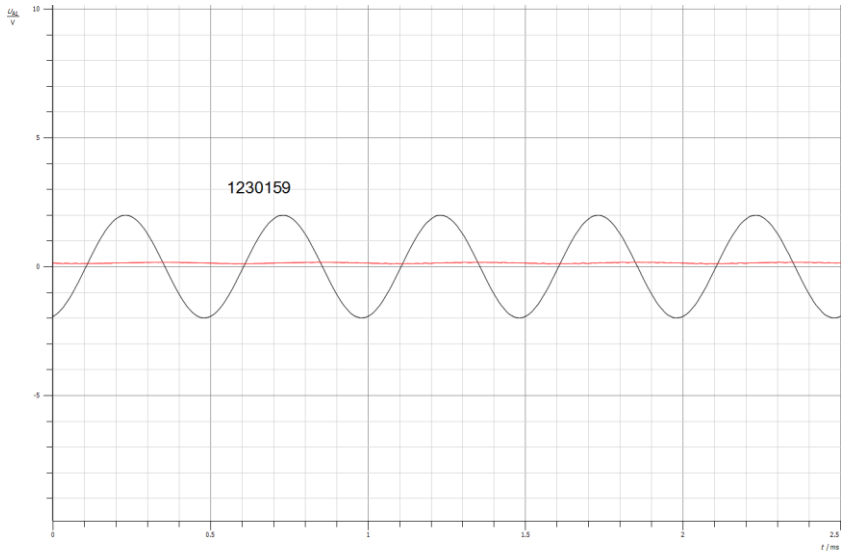


Figure 28 non-coherent demodulation in time domain

Output in frequency domain

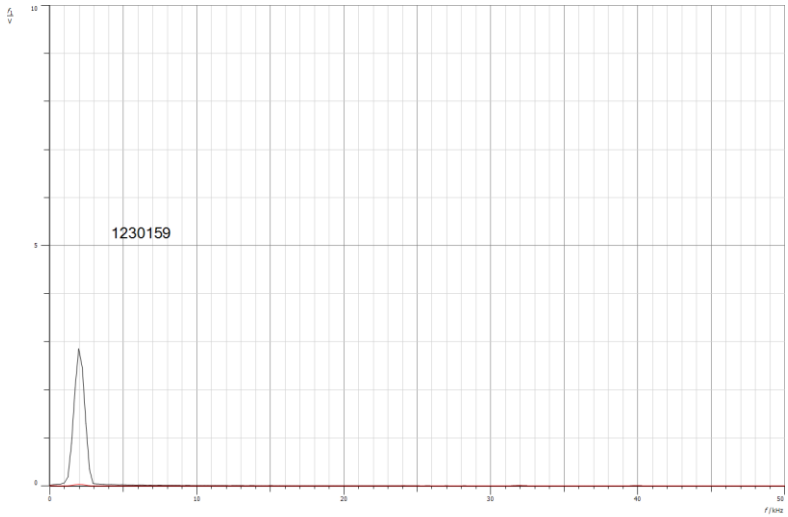


Figure 29 non-coherent demodulation in frequency domain

In non-coherent demodulation the phase will be 90, and the message signal will disappear, because the cos (90) =0, so “Am*0” → the signal will be zero.

Conclusion:

This experiment provided a solid understanding of amplitude modulation (AM) and demodulation techniques. Modulation is classified into three types based on the modulation index: under modulation, normal modulation, and over modulation. Among these, normal modulation (modulation index = 1) gave the best performance, resulting in minimal distortion and accurate signal recovery. In contrast, under modulation (modulation index < 1) and over modulation (modulation index > 1) can cause poor or failed demodulation.

Modulation is essential for transmitting signals effectively in different environments, while demodulation allows the receiver to recover the original message. The carrier frequency determines the position of the signal on the frequency spectrum, while the bandwidth of an AM signal is equal to twice the highest frequency of the message signal ($2f_m$). The modulation index depends only on the amplitudes of the message and carrier signals, not their frequencies.

In this experiment, two demodulation methods coherent and non-coherent—were implemented. Filters were also used during demodulation to extract the original message signal from the modulated wave, and the circuit's time constant played a key role in ensuring accurate signal recovery.

Finally, AM offers several advantages, including simple transmitter and receiver design, lower cost, and the ability to cover long distances, making it suitable for broadcasting applications. However, it also has disadvantages such as poor noise immunity, lower sound quality compared to other modulation methods, and inefficient power usage due to the constant transmission of the carrier signal

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